ENEL 525 – Machine Learning for Engineers

Course Project Description

Fall Session 2026

Item	Details
Course	ENEL 525 – Machine Learning for Engineers, University of Calgary
Project weight	20% of the final course grade
Team size	Groups of 2
Final deadline	December 04, 2026

1. Overview

In this project you will design, build, and critically evaluate a machine learning solution to a problem of your own choosing. The application area is entirely open: sports analytics, finance, robotics, healthcare, energy systems, transportation, agriculture, music/acoustics, natural language, or anything else that interests you. What matters is not the domain but the quality of your machine learning understanding as Engineers: how clearly you define the problem, how carefully you prepare and handle data (This is a very important step!), how well you justify your modelling choices, and how honestly you evaluate and interpret your results.

The project is your opportunity to apply the techniques covered in the course (for example regression, classification, neural networks, regularization, model selection, and unsupervised learning) to a problem that matters to you, and to practice communicating technical work the way a professional engineer would.

2. Learning Objectives

By completing this project, you should be able to:

- Formulate a real-world problem as a well-posed machine learning task with a measurable objective.
- Acquire, explore, clean, and prepare data while identifying and mitigating common pitfalls such as leakage, imbalance, and bias.
- Select, implement, and tune appropriate models, and justify these choices against a meaningful baseline.
- Evaluate models using suitable metrics and validation strategies and interpret the results critically.
- Discuss limitations, failure modes, and ethical considerations of your solution.
- Communicate technical results clearly in writing and in a short presentation.

3. Choosing a Project

You may choose any problem for which machine learning is a reasonable approach and for which suitable data is available. Some illustrative directions are listed below; you are encouraged to propose your own.

Domain	Example project ideas
Sports	Predict match outcomes or player performance; classify play types from tracking data; estimate injury risk from workload data.
Finance	Forecast volatility or demand; detect fraudulent transactions; credit-risk scoring.
Robotics	Learn a control policy in simulation; object detection for grasping; terrain classification from sensor data.
Energy and environment	Short-term load or solar-output forecasting; air-quality prediction; fault detection in equipment.
Health and biomedical	Classify signals (ECG, EEG) or images; predict outcomes from public clinical datasets.
Language, audio, vision	Text classification or summarization; music genre recognition; image segmentation for a specialized acoustic task.

Project constraints

- The project must involve learning from data; a purely rule-based or hand-coded solution is not acceptable.
- The scope should be achievable within the project timeline. A well-executed, focused project is preferred over an overly ambitious one that is incomplete.
- Data must be obtainable legally and ethically (public datasets, simulations, or data you are authorized to use). Do not use personally identifiable or confidential data without explicit permission.
- Projects that simply re-run a tutorial or a published notebook without meaningful extension will not receive a passing mark.

4. Technical Requirements

Every project must address the following components. The rubric in Section 7 shows how each is assessed.

4.1 Problem definition and motivation

State the problem, why it matters, who would benefit from a solution, and what a successful outcome might look like. Define the learning task (e.g., regression, classification, clustering, sequence prediction, reinforcement learning) and the quantitative success criteria.

4.2 Data

- Describe the data source, size, features, labels, and how it was collected.
- Perform exploratory data analysis (EDA) with informative visualizations and summary statistics.
- Document all preprocessing steps (cleaning, normalization, feature engineering, handling missing values, class imbalance) and justify them.
- Split the data properly into training, validation, and test sets (or use cross-validation), and explain how you prevented data leakage. For time-series data, respect temporal ordering.

4.3 Methodology

- Establish at least one simple baseline (e.g., a naive predictor or a linear/logistic model).
- Implement at least two distinct modelling approaches in addition to the baseline, with at least one that goes beyond a default library call (e.g., a neural network you design, a carefully engineered feature pipeline, or a custom loss or training procedure).
- Perform systematic hyperparameter tuning using the validation set only, and report the search procedure.
- Justify each modelling choice in terms of the data and the problem.

4.4 Evaluation and analysis

- Select metrics appropriate for the task and the data (e.g., F1, ROC-AUC, or precision/recall for imbalanced classification; RMSE or MAE for regression) and explain why.
- Report final results on the held-out test set, compared against the baseline, ideally with uncertainty estimates (e.g., repeated runs or confidence intervals).
- Include at least one of: ablation study, error analysis, learning curves, feature importance or interpretability analysis.
- Discuss limitations, failure cases, generalization, and potential bias or ethical concerns. Describe what you would do next with more time or data.

4.5 Code and reproducibility

Your code must run end-to-end from a clean environment. Use a Jupyter/Colab notebook or a well-structured repository, include a README with setup instructions, a requirements file, random seeds, and clear instructions for obtaining the data.

5. Deliverables and Deadlines

Deliverable	Description	Format / length	Deadlines
Project proposal	Title, problem statement, motivation, proposed dataset, planned approach, evaluation metrics, and team members. Feedback will be provided.	1–2 pages, PDF	Submission: October 26, 2026 Feedback: October 30, 2026
Final report	Complete write-up covering introduction, related work, data, methods, results, discussion, conclusions, and references.	6–8 pages in IEEE two-column format, PDF	Submission: December 04, 2026
Code	Fully reproducible implementation with README and dependency list.	GitHub repository or Colab notebook link	Submission: December 04, 2026
Presentation	Short talk summarizing the problem, approach, key results, and lessons learned, followed by questions.	12-15 minutes + Q&A	November 27 & December 04, 2026
Contribution statement (teams only)	Description of each member's contributions.	This can be added at the end of the report (Extra 1 page allowed)	Submission: December 04, 2026

6. Policies and Expectations

Academic integrity

All submitted work must be your own (or your team's). Cite every dataset, paper, library, code snippet, picture and other external resource you use. Reusing code or ideas without attribution is academic misconduct under University of Calgary policy.

Use of AI tools

If AI-assisted tools are used, you must disclose how they were used, and you remain fully responsible for understanding, verifying, and being able to explain every part of your submission. You may be asked to explain your work in the presentation Q&A.

Teamwork

Teams are expected to share the workload fairly. A contribution statement is required, and individual marks may be adjusted when contributions are clearly unequal.

Ethics and data

Consider privacy, fairness, and the potential for misuse of your solution. Do not collect or scrape data in violation of a website's terms of use or applicable privacy law.

Late submissions

There will be:

- 10% deduction for late submission (Proposal or final report+presentation)
- 50% deduction for no show for the presentation.
- 100% deduction for no project submission and no presentation.

7. Evaluation Rubric

Your project will be graded out of 100 using the rubric below. Each criterion is scored by performance level; the score is multiplied by the criterion weight. Individual marks for team projects may be adjusted based on the contribution statement and presentation Q&A.

Criterion	Weight	Excellent (9-10)	Proficient (7-8)	Developing (4-6)	Insufficient (< 4)
Project Proposal: problem statement, motivation, proposed dataset, planned approach, proposed evaluation metrics, and team members.	**20%**	Problem is clearly and precisely stated, well-motivated, and appropriately scoped. Learning task and quantitative success criteria are explicit.	Problem is clear and motivated; minor gaps in scope or success criteria.	Problem is vaguely stated or weakly motivated; success criteria are unclear or incomplete.	Problem is unclear, unsuitable for ML, or missing.
Data collection, exploration, and preprocessing	**10%**	Data source well documented. Thorough, insightful exploratory data analysis. All preprocessing justified. Correct splitting with explicit leakage prevention.	Data adequately described and explored. Preprocessing mostly justified. Splitting is correct.	Limited exploratory data analysis or poorly justified preprocessing. Questionable splitting or possible leakage.	Little or no data analysis. Improper splitting or evident leakage invalidates results.
Methodology and model development	**10%**	Meaningful baseline plus two or more well-chosen approaches, including one beyond a default library call. Choices clearly justified. Systematic tuning using validation data only.	Baseline and two approaches implemented correctly. Choices mostly justified. Tuning performed but limited.	Baseline or comparison missing, or models used without justification. Minimal or ad hoc tuning.	Methods are inappropriate, incorrectly implemented, or no baseline is provided.
Experimental evaluation	**10%**	Appropriate metrics well justified. Rigorous test-set results compared with baseline, with uncertainty estimates. Conclusions fully supported by evidence.	Appropriate metrics and sound test-set evaluation. Comparison with baseline is clear; limited treatment of uncertainty.	Metrics partially appropriate. Evaluation incomplete or comparisons unclear. Some conclusions not well supported.	Inappropriate metrics or no valid held-out evaluation. Claims unsupported.

Criterion	Weight	Excellent (9-10)	Proficient (7-8)	Developing (4-6)	Insufficient (< 4)
Analysis, discussion, limitations and ethics	**10%**	Insightful analysis (ablation, error analysis, interpretability, or learning curves). Transparent discussion of limitations, failure modes, bias, and ethics. Thoughtful future work.	Good analysis with at least one deeper investigation. Limitations and ethical aspects addressed.	Superficial analysis. Limitations or ethical considerations mentioned only briefly.	Results reported without interpretation. No discussion of limitations.
Report quality	**10%**	Well organized, concise, and professional. Figures and tables are clear and informative. Properly cited and in the required format.	Clear and organized with minor issues in writing, figures, citations, or format.	Disorganized or unclear in places. Weak figures. Incomplete citations or format.	Hard to follow. Missing sections or citations.
Code and reproducibility	**10%**	Clean, documented code runs end-to-end from a clean environment. README, dependencies, and seeds provided.	Code runs with minor effort. README and dependencies provided.	Code runs only with significant fixes. Documentation is sparse.	Code is missing or does not run.
Presentation and Q&A (Detailed rubrics will be shared)	**20%**	Engaging, clear, well-paced, and within time. Excellent visuals. Confident, accurate answers that show deep understanding.	Clear and organized with good visuals. Questions answered adequately.	Unclear structure or timing issues. Difficulty answering questions.	Disorganized or incomplete. Cannot explain the work.
Total	**100%**				

Please reach out to me at iqra.jawaid@ucalgary.ca for any question or concerns.